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COURSE CODE:DSA0404

CO3-AT1 CASE STUDY

1. **Detecting an Unusual Customer A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400. A particular customer spends ₹3,200.**

    1. Calculate the customer's z-score.

    2. Is the customer's spending unusually high compared with the average?

    3. What would a negative z-score mean in this context?

    4. If another customer has a z-score of −1.5, what does it tell you?

Given:

- Average spending (mean) $\mu = ₹2000$
- Standard deviation $\sigma = ₹400$
- Customer spending $x = ₹3200$

We use the z-score formula:

$$z = \frac{(1.2) - (0)}{1} = 1.2$$

$$\Phi(z) \approx 88.5\%$$

x =1.2

μ =0.0

σ =1.0

1. Calculate the customer's z-score

Substitute the values:

$$z = \frac{3200 - 2000}{400} = \frac{1200}{400} = 3$$

Z-score = 3

2. A z-score of 3 means the customer spends 3 standard deviations above the average.

In statistics:

- About 99.7% of values in a normal distribution lie within ±3 standard deviations.
- A value at z = 3 is very rare and considered unusually high.

Yes, the customer's spending is unusually high compared with the average.

3. A negative z-score means the customer spends less than the average.

Examples:

- $z = -1$→ spends 1 standard deviation below average.
- $z = -2$→ spends 2 standard deviations below average.

So, a negative z-score indicates below-average spending.

4. A z-score of −1.5 means the customer spends 1.5 standard deviations below the average.

Calculate the spending amount:

$$x = \mu + z\sigma$$

$$x = 2000 + (-1.5)(400)$$

$$x = 2000 - 600 = ₹1400$$

This customer spends ₹1,400.

2. **Delivery Time Analysis A delivery company records delivery times (in minutes): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60**
    1. **Calculate the mean and median.**
    2. **Find Q1 and Q3.**
    3. **Calculate the IQR.**
    4. **Check for outliers using the 1.5 × IQR rule.**
    5. **Explain whether the mean or median gives a better description of the typical delivery time.**

Let the delivery times (in minutes) be:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

The data are already arranged in ascending order.

Number of observations: n = 10

1. **Calculate the Mean and Median**

Mean:

$$\bar{x} = \frac{2+4+7}{3} = \frac{13}{3} = 4\,\frac{1}{3}$$

X1=2

X2=4

X3=7

Sum of all delivery times:

$$25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

$$\text{Mean} = \frac{326}{10} = 32.6$$

Mean = 32.6 minutes

Median:

Since there are 10 values (even number), the median is the average of the 5th and 6th values.

5th value = 30

6th value = 30

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Median = 30 minutes

2. Find Q1 and Q3

Split the data into two halves.

Lower half

25, 27, 28, 29, 30

Median of lower half = 28

So, Q1 = 28

Upper half

30, 31, 32, 34, 60

Median of upper half = 32

So, Q3 = 32

Q1 = 28 minutes

Q3 = 32 minutes

3. Calculate the IQR

$$\text{IQR} = Q_3 - Q_1$$

$$32 - 28 = 4$$

IQR = 4 minutes

4. Check for Outliers (1.5 × IQR Rule)

Lower fence

$$Q_1 - 1.5(\text{IQR}) = 28 - 1.5(4) = 28 - 6 = 22$$

Upper fence

$$Q_3 + 1.5(\text{IQR}) = 32 + 1.5(4) = 32 + 6 = 38$$

Any value:

- less than 22, or
- greater than 38

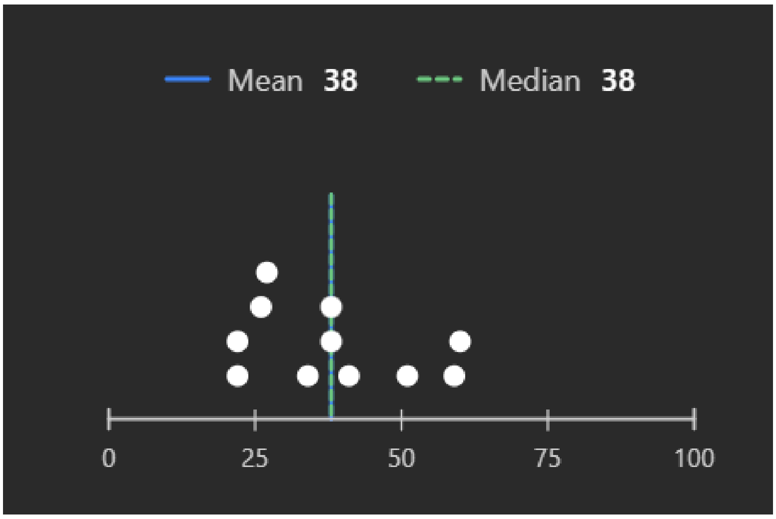
is an outlier.

From the data, 60 > 38.

60 minutes is an outlier.

No lower outlier exists.

5. Which is Better: Mean or Median?



- The value 60 minutes is unusually high compared with the other delivery times (25–34 minutes).
- This outlier increases the mean from around 30 to 32.6 minutes.
- The median remains 30 minutes, which is close to the center of the majority of deliveries.

3. **Two Cities' Temperatures**

Given Data

- **City A: 28, 29, 30, 30, 31, 30, 32**
- **City B: 20, 25, 30, 35, 30, 25, 45**

1. Calculation of Mean Temperature

Formula

Mean = (Sum of all values) / (Number of values)

City A

Step 1: Find the sum

28 + 29 + 30 + 30 + 31 + 30 + 32 = **210**

Step 2: Count the values

n = **7**

Step 3: Calculate the mean

Mean = 210 / 7 = **30°C**

Mean temperature of City A = 30°C

City B

Step 1: Find the sum

20 + 25 + 30 + 35 + 30 + 25 + 45 = **210**

Step 2: Count the values

n = **7**

Step 3: Calculate the mean

Mean = 210 / 7 = **30°C**

Mean temperature of City B = 30°C

2. Calculation of Variance

Formula

Variance = $\Sigma(x - \bar{x})^2 / n$

where $\bar{x}$ = 30°C

City A

x	x − 30	(x − 30)²
28	−2	4
29	−1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Step 1: Sum of squared differences

4 + 1 + 0 + 0 + 1 + 0 + 4 = 10

Step 2: Divide by n

Variance = 10 / 7 = 1.43

Variance of City A = 1.43°C²

City B

x	x − 30	(x − 30)²
20	−10	100
25	−5	25
30	0	0
35	5	25
30	0	0
25	−5	25
45	15	225

Step 1: Sum of squared differences

100 + 25 + 0 + 25 + 0 + 25 + 225 = 400

Step 2: Divide by n

Variance = 400 / 7 = 57.14

Variance of City B = 57.14°C^2

3. Comparison of Temperature Variability

City	Variance
City A	1.43
City B	57.14

Since **57.14 > 1.43, City B has greater temperature variability.**

Interpretation

- Both cities have the same average temperature (30°C).
- City A's temperatures are closely clustered around the mean, so its variance is very small.
- City B's temperatures are spread widely from 20°C to 45°C, resulting in a much larger variance.

4. Data Center CPU Usage A data scientist records CPU usage percentages: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.
2. Calculate the range.
3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.

Given Data

CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

The data are already arranged in ascending order.

Number of observations: n = 10

1. Mean, Median, and Mode

Mean

Sum of all values:

40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513

Mean = 513 / 10 = 51.3%

Median

Since n = 10 (even), the median is the average of the 5th and 6th values.

5th value = 46

6th value = 48

Median = (46 + 48) / 2 = 47%

Mode

The value occurring most frequently is 45.

Mode = 45%

2. Range

Range = Maximum − Minimum

Range = 90 − 40 = **50%**

3. Quartiles and Interquartile Range (IQR)

Lower half

40, 42, 45, 45, 46

Median of lower half = **45**

Q1 = **45%**

Upper half

48, 50, 52, 55, 90

Median of upper half = **52**

Q3 = **52%**

Interquartile Range

IQR = Q3 − Q1

IQR = 52 − 45 = **7%**

4. Identification of Possible Outliers

Lower fence

Q1 − 1.5(IQR) = 45 − 1.5(7) = 45 − 10.5 = **34.5**

Upper fence

Q3 + 1.5(IQR) = 52 + 1.5(7) = 52 + 10.5 = **62.5**

Any value less than 34.5 or greater than 62.5 is an outlier.

In the data, **90 > 62.5**, so **90% is a possible outlier**.

No lower outlier exists.

5. Direction of Skewness

Compare the measures of central tendency:

- Mean = **51.3**
- Median = **47**
- Mode = **45**

Since **Mean > Median > Mode**, the distribution is **positively skewed (right-skewed)**.

The large value 90 creates a long tail on the right side of the distribution.

5. Comparing Two Datasets Two data analysts receive the following datasets: Dataset A: 10, 20, 30, 40, 50 Dataset B: 28, 29, 30, 31, 32 1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why mean alone is not enough to compare datasets

Given Data

- **Dataset A:** 10, 20, 30, 40, 50
- **Dataset B:** 28, 29, 30, 31, 32

Number of observations in each dataset: **n = 5**

1. Mean of Both Datasets

Formula

Mean = (Sum of all values) / n

Dataset A

Sum = 10 + 20 + 30 + 40 + 50 = **150**

Mean = 150 / 5 = **30**

Dataset B

Sum = 28 + 29 + 30 + 31 + 32 = **150**

Mean = 150 / 5 = **30**

Result

- **Mean of Dataset A = 30**
- **Mean of Dataset B = 30**

Both datasets have the same mean.

2. Variance of Both Datasets

Formula

Variance = $\Sigma(x - \bar{x})^2 / n$

where $\bar{x} = 30$.

Dataset A

x	x – 30	(x – 30)²
10	−20	400
20	−10	100
30	0	0
40	10	100
50	20	400

Sum of squared deviations = 1000

Variance = 1000 / 5 = 200

Variance of Dataset A = 200

Dataset B

x	x − 30	$(x - 30)^2$
28	−2	4
29	−1	1
30	0	0
31	1	1
32	2	4

Sum of squared deviations = 10

Variance = 10 / 5 = 2

Variance of Dataset B = 2

3. Standard Deviation of Both Datasets

Formula

Standard Deviation = √Variance

Dataset A

SD = √200 ≈ 14.14

Dataset B

SD = √2 ≈ 1.41

Result

- **Standard deviation of Dataset A ≈ 14.14**
- **Standard deviation of Dataset B ≈ 1.41**

4. Which Dataset Has Greater Spread?

Compare the variances or standard deviations:

Dataset	Variance	Standard Deviation
A	200	14.14
B	2	1.41

Dataset A has a much larger variance and standard deviation.

Conclusion

Dataset A has greater spread.

5. Which Model Appears More Consistent?

If the datasets represent model errors:

- Dataset A errors vary widely from 10 to 50.
- Dataset B errors stay close to 30.

A model is considered more consistent when its errors show less variability.

Conclusion

The model represented by Dataset B is more consistent.

6. Why Mean Alone Is Not Enough

Although both datasets have the same mean (30), their distributions are very different.

- Dataset A is widely spread around the mean.
- Dataset B is tightly clustered around the mean.

The mean only tells us the central value; it does not describe how spread out the data are. Measures such as variance and standard deviation are needed to understand consistency, reliability, and variability.

For example, two models can have the same average error, but one model may produce highly unstable predictions while the other produces very stable predictions.